The model we provide shows trends and will never be able to tackle, at the local scale of these sub-catchments, the random processes associated with stochastic weather events. Together with the hazard assessment, we provide the inventory of the deep-seated landslides and highlight the susceptibility to ground deformations. Since we do not have measure of the ground deformation rates, we cannot provide a hazard scenario. However, as observed in other regions, and notably in the region of Bukavu in DRC that is close to Rwanda, ground deformations in such landslides can be pervasive (Nobile et al., 2018; Dille et al., 2021). As such, on a short time scale, independent of the weather conditions, disturbances due to existing deep-seated landslides are potentially higher. This needs further investigation.

2.6 Step 4: Exposure and vulnerability

2.6.1 Objectives

The goal of exposure and vulnerability assessment is to build an exposure database and assess the vulnerability of the elements at risk to landslides. Seeing we did not receive any vulnerability databases produced by other parties to assess the elements at risk based on previous landslides, we carried out a spatial overlap of hazard zonation and elements at risk. Such an approach using a simple exposure analysis is common in the literature as acquiring detailed vulnerability information remains generally a challenge, which would require significant research investment that is not compatible with the duration of this study (Glade et al., 2006; Corominas et al.,2014).

2.6.2 Results

Results were produced from a desktop exercise by spatially overlapping the hazard zonation (landslide rate) developed in Step 3 (Figure 8) with the different land use categories. Results are presented for two scenarios summarised in tables 8 and 9 below: -

Elements at risk for the current land use situation;

- Elements at risk for the projected land use master plan 2050.
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Landslide rate (m^2 /year/km^2)
1
18
90
365